

Genetics in the news



The strawberry genome: a complicated past and promising future | Horticulture Research

Within the juicy red flesh of the commercial strawberry lies a deep history that spans at least three continents and hundreds of thousands of years. The genetic heritage of this prized dessert fruit was crafted by nature, with its modern improvement driven by the indigenous Mapuche people of South America, a seafaring French spy, and a plant-loving teenager that gathered fruits to present to a king. The many stories that punctuate the history of the modern strawberry relate to its contemporary cultivation as well as origins that span a significant portion of the globe^{1,2}. The evidence of this rich history is locked away within the chromosomes of every cell, each archiving a complicated story that speaks both of strawberry's historical origins and genetic potential for future production.

The complete genetic storybook of the commercial strawberry has remained a secret to researchers, until now.

Genetics in the news

[This Mutant Crayfish Clones Itself, and It's Taking Over Europe](#)



Triploid

- Therefore larger in size
- Possibly caused by dispermy in crowded “petshop” aquarium
- Parthenogenic
- Can reproduce without mating



The Molecular Basis of Inheritance

From Chromosomes to Genomes

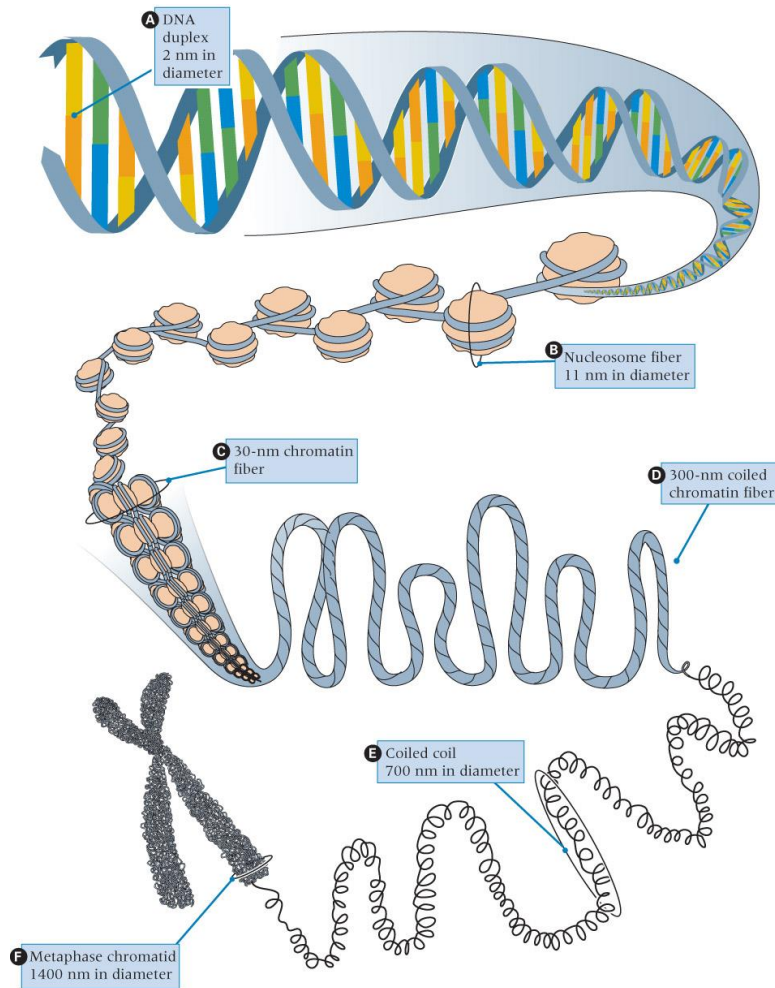
BIOL 202

Prof. Laura Nilson, Dept of Biology

Office hours

- Fridays 1-2pm (zoom or Stewart Biology, N5/8)
<https://mcgill.zoom.us/j/81407275167?pwd=8Cgm0RPnWQRLOdaCNFyXnS3M8amQUj.1>
- By appointment
- Before or after class, right here in Leacock 132

Here's an overview of this section



- Large scale changes in chromosome number or structure
- Transposable elements and genome surveillance
- DNA mutations: causes and consequences
- Understanding the genetic code: genetics, bioinformatics, and functional genomics
- Your input: Other topics of interest???

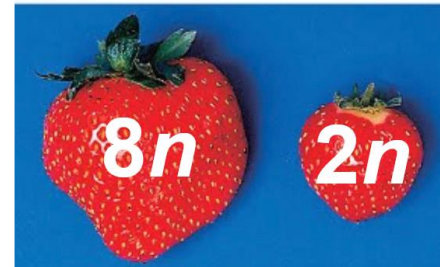
Autopolyploid

An individual that has multiple chromosome sets originating from within one species



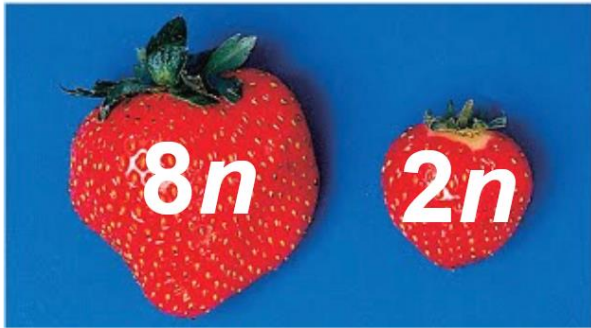
Allopolyploid

An individual that has multiple chromosome sets originating from two or more different species



[The strawberry genome: a complicated past and promising future | Horticulture Research](#)

Octoploid strawberry plants are allopolyploids that arose through natural sequential hybridization events involving different diploid strawberry species



[Strawberries Have 8 Sets of Chromosomes to Thank for Their Survival](#) [| Scientific American](#)

In 2019, my colleagues and I [published the first high-quality genome of the strawberry plant](#), which revealed the octoploid genome arose by a stepwise process. At some point over a million years ago, two ancient diploid species hybridized and produced a now-extinct plant species with four sets of chromosomes; that species hybridized with a third diploid species, resulting in six sets of chromosomes, and then with a fourth diploid species, resulting in eight sets of chromosomes. This ancient wild octoploid then spread throughout the Western Hemisphere, splitting into two species that European colonists collected in the 18th century; those plants underwent a final hybridization event in continental Europe around 300 years ago to create the strawberry you know and love in your grocery store or garden.

Origin of three allopolyploid species of *Brassica*

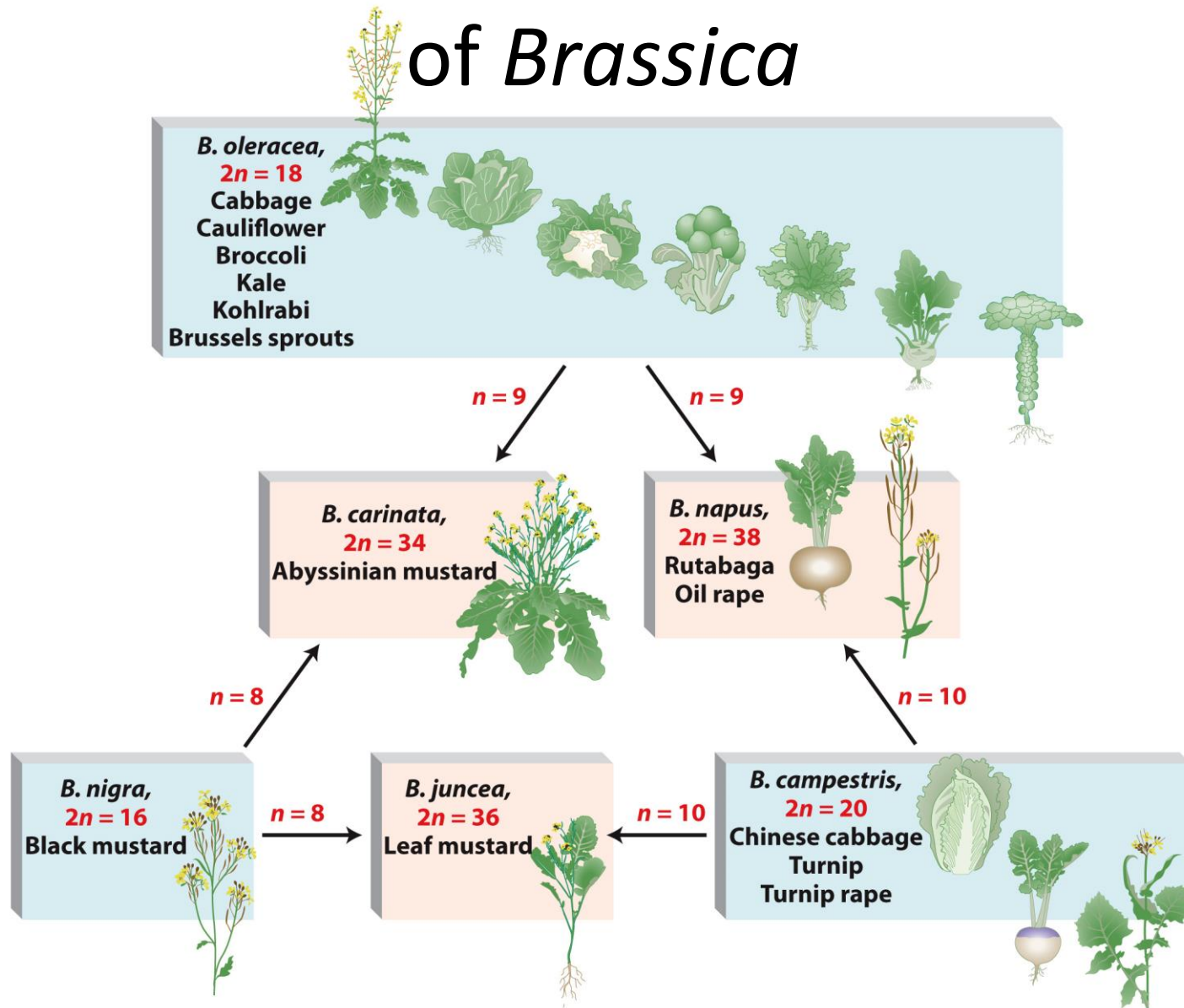
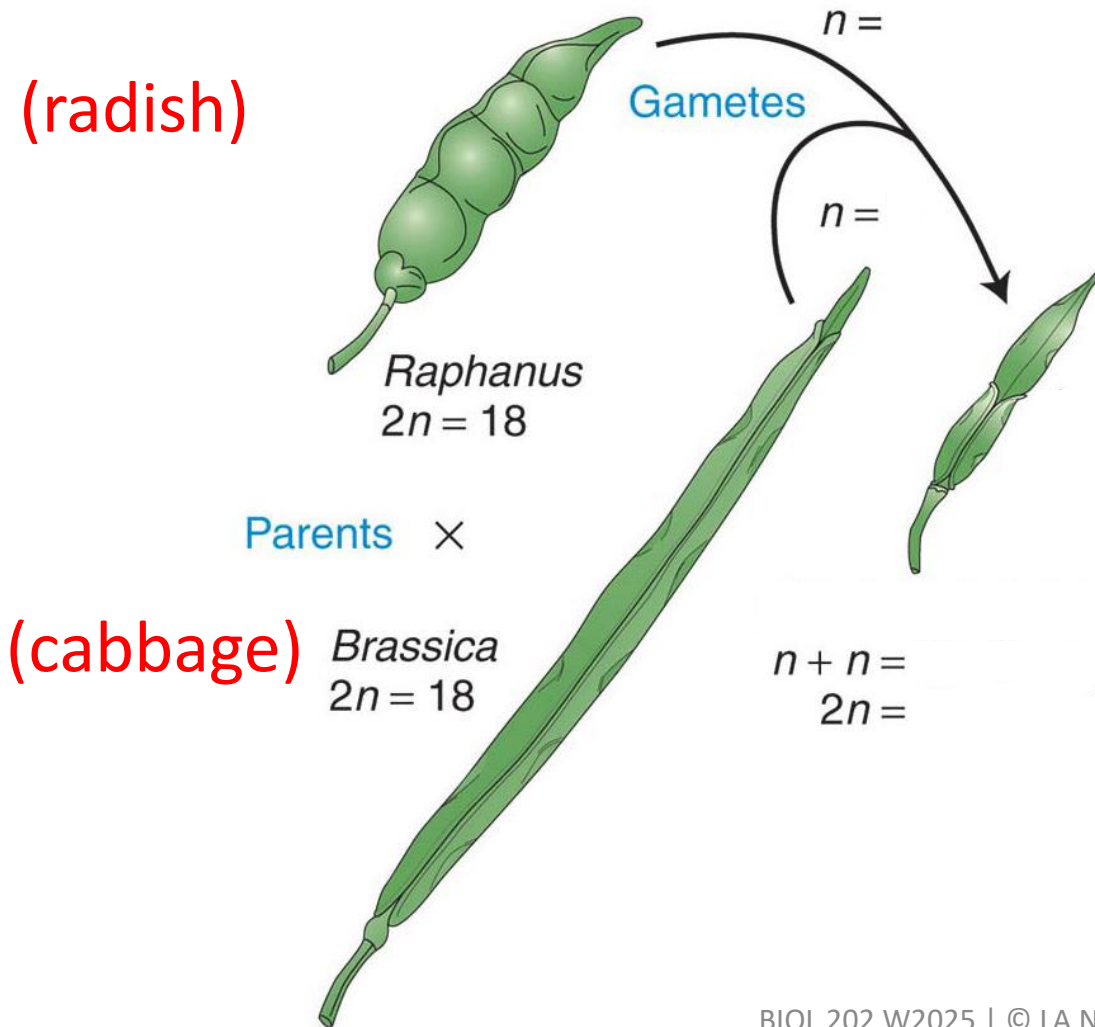


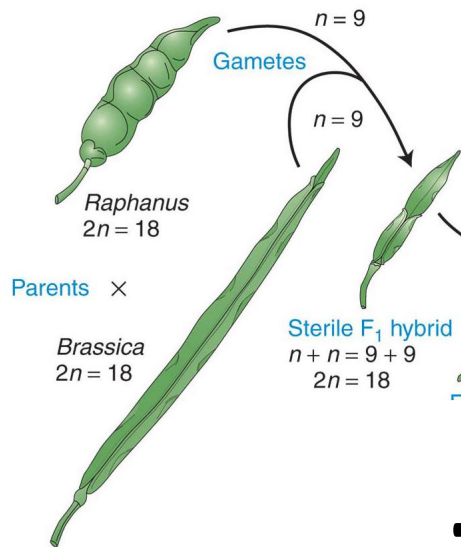
Figure 17-8

Introduction to Genetic Analysis, Eleventh Edition

© 2015 W. H. Freeman and Company

Generation of an allopolyploid: a hybrid of two species

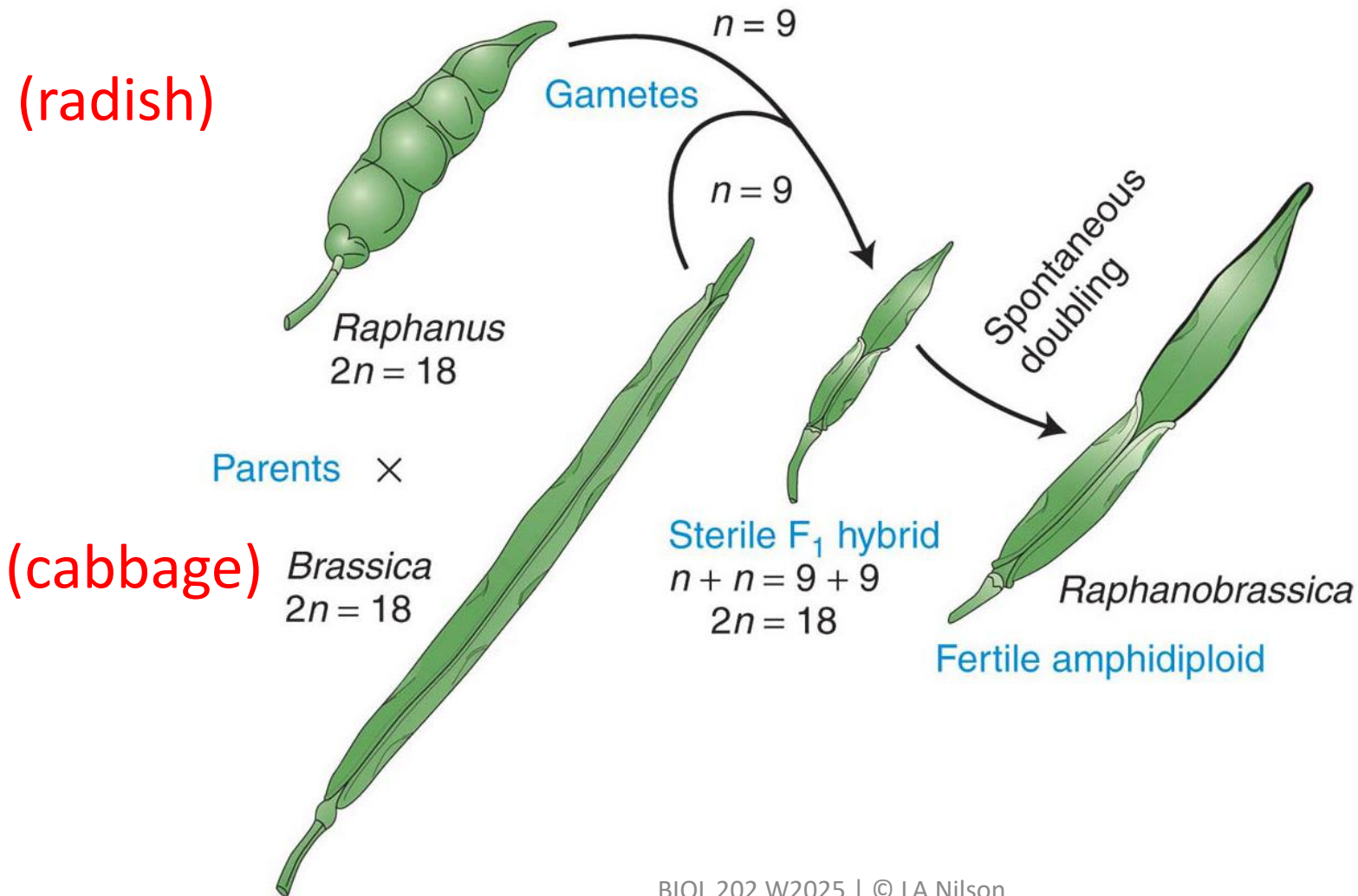




© 2015 W.H. Freeman and Company

Think Break

Generation of an allopolyploid



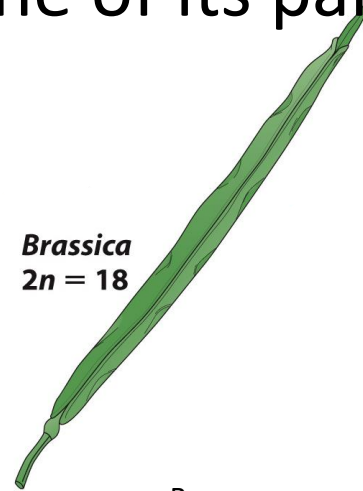
Crossing an allopolyploid back to one of its parent species



Raphanobrassica
Fertile amphidiploid
 $2n + 2n = 18 + 18$
 $4n = 36$
Parent: $2n^R + 2n^B$
Gametes: $n^R + n^B$

X

Brassica
 $2n = 18$



Parent: $2n^B$
Gametes: n^B



F1? F1: $n^R + n^B + n^B$



F2?

Interpretation? These two plants can no longer breed with each other → new species

Origin of three allopolyploid species of *Brassica*

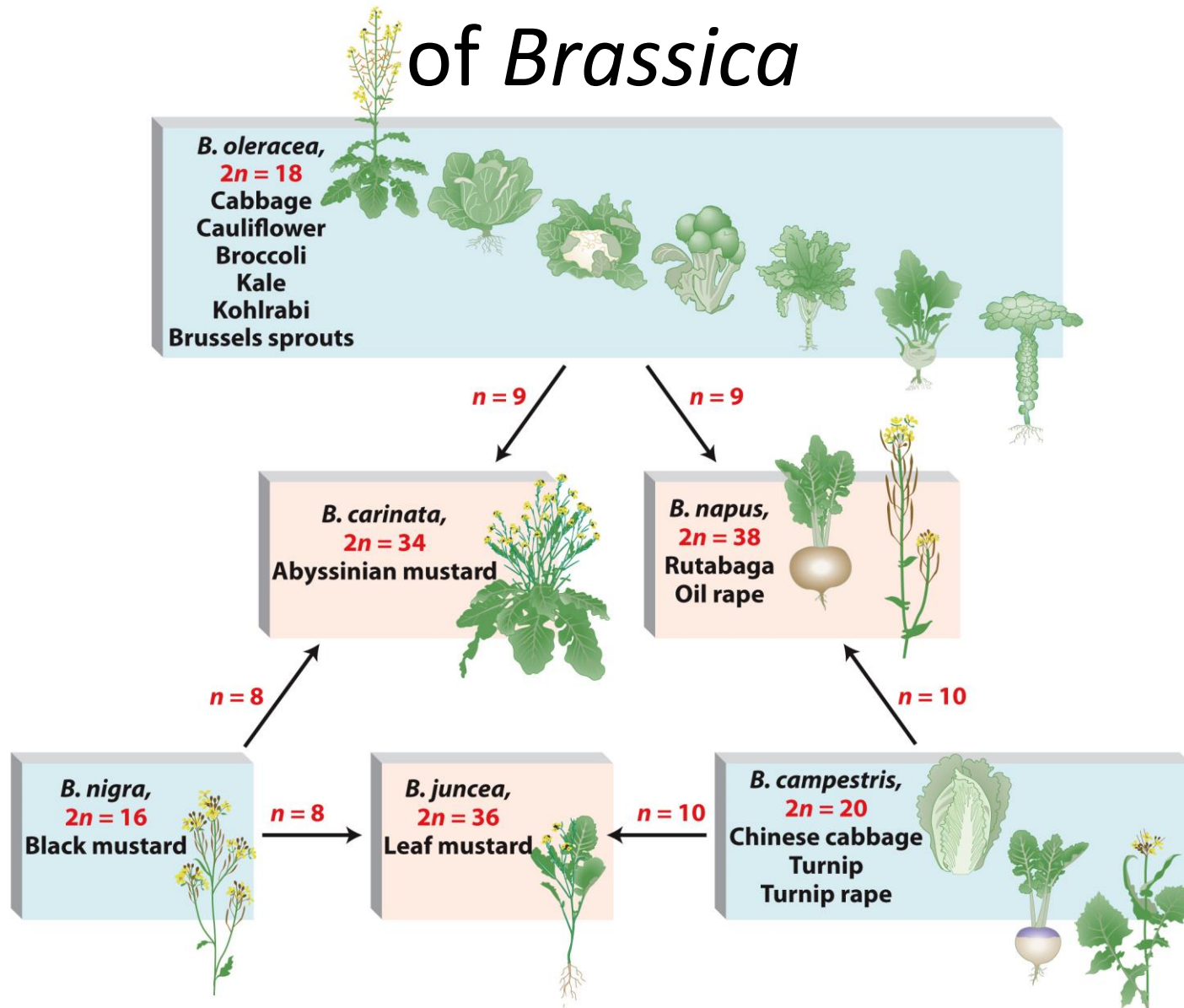


Figure 17-8

Introduction to Genetic Analysis, Eleventh Edition

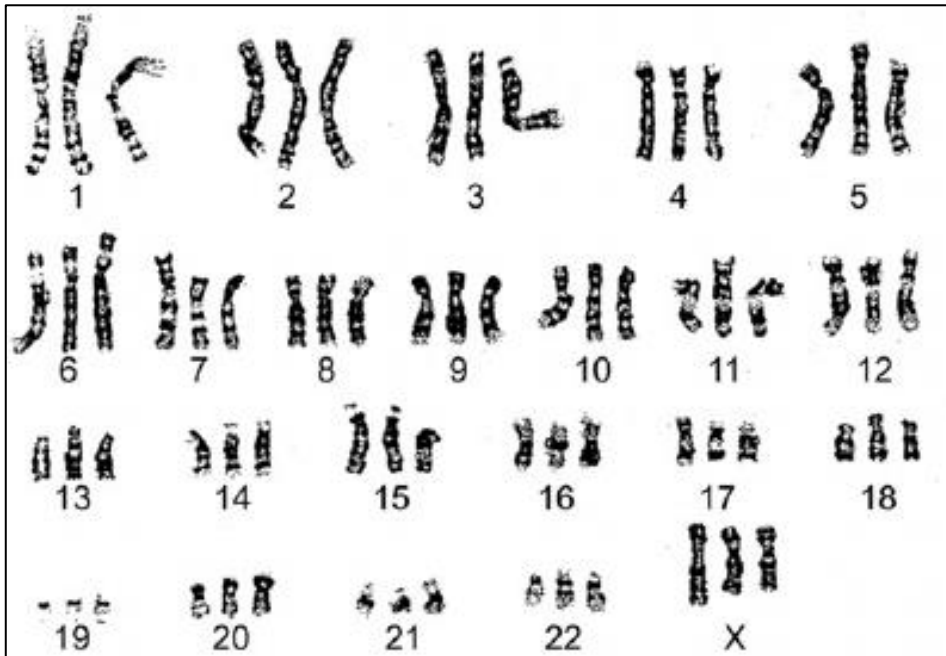
© 2015 W. H. Freeman and Company

Large Scale Chromosomal Changes

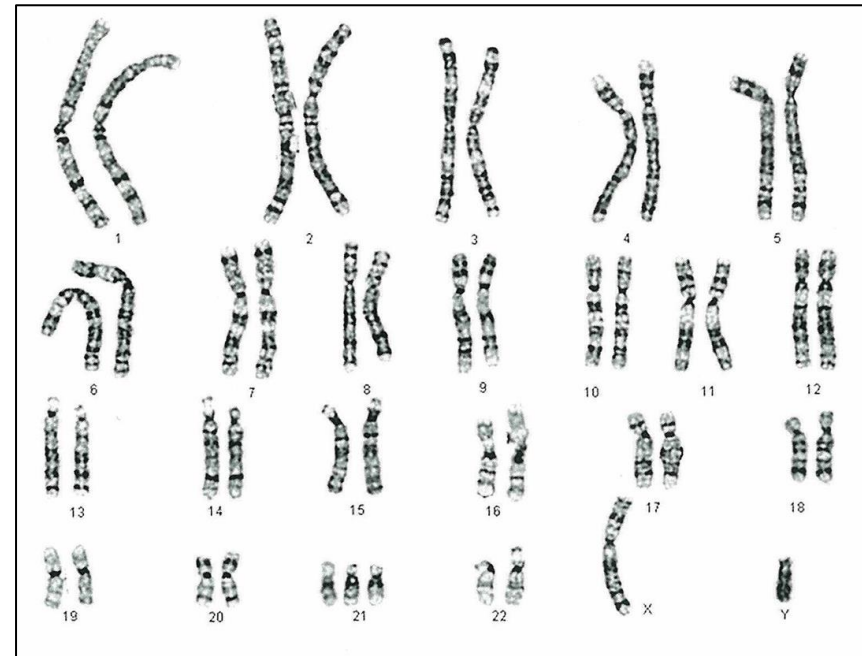
Changes in Chromosome Number
ANEUPLOIDY

Polyploidy vs. Aneuploidy

Polyploid ; triploid ; $3n$ (EUPLOID)



Trisomic ; $2n+1$ (ANEUPLOID)



Effects on the soma
Effects on the germline
How does aneuploidy arise?

Trisomy

$2n+1$

viable trisomies

XXX:
normal

XYY:
normal

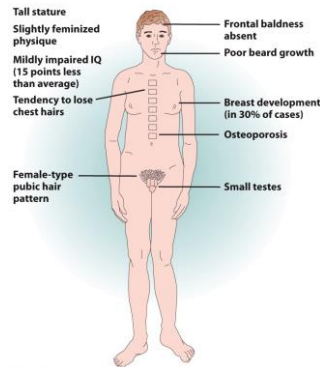


Figure 17-15
Introduction to Genetic Analysis, Eleventh Edition
© 2015 W. H. Freeman and Company

XXY:
Klinefelter
syndrome

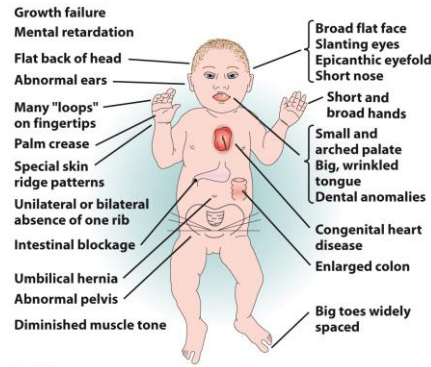


Figure 17-16
Introduction to Genetic Analysis, Eleventh Edition
© 2015 W. H. Freeman and Company

trisomy 21:
Down syndrome

other trisomies

- Trisomy 13, Trisomy 18: non-viable, usually die in infancy
- All others: non-viable, die in utero

$2n-1$

viable monosomy

XO: Turner syndrome

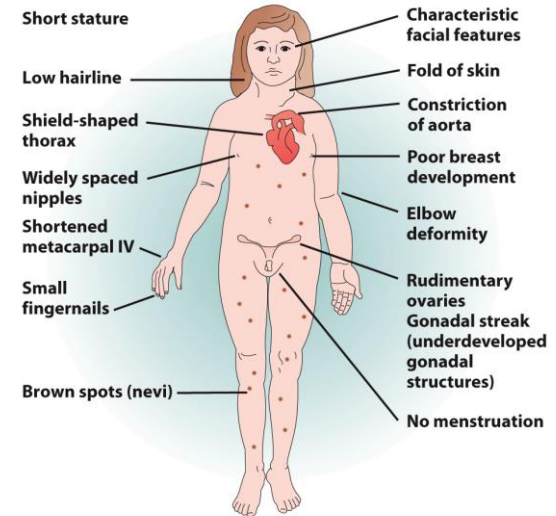
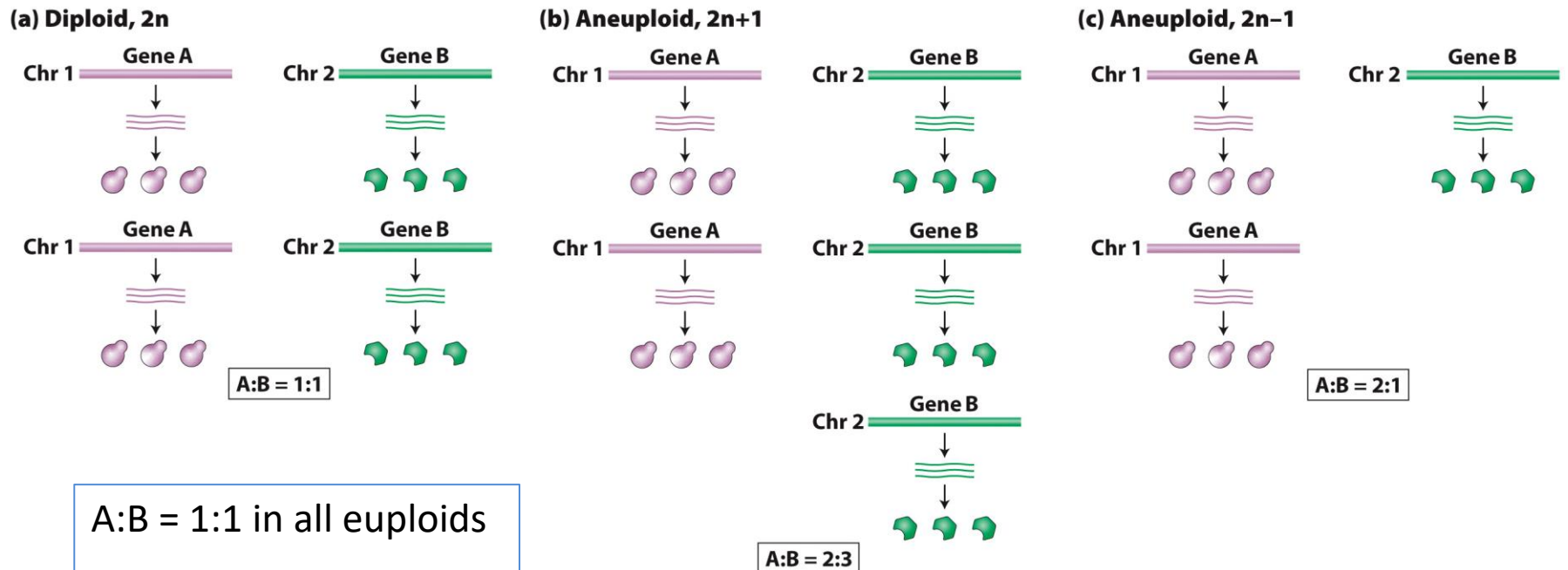


Figure 17-13
Introduction to Genetic Analysis, Eleventh Edition
© 2015 W. H. Freeman and Company

Aneuploidy affects the balance of gene dosage in a cell

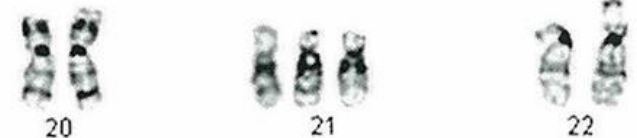


A:B = 1:1 in all euploids

e.g. triploid = 3:3 = 1:1

Figure 17-17

Introduction to Genetic Analysis, Twelfth Edition
© 2020 W. H. Freeman and Company



Human trisomy 21:

- Only trisomy to routinely survive to adulthood
- Phenotypes of partial trisomy 21s help identify which regions contribute to Down Syndrome

The trisomics of *Datura* (Jimsonweed)

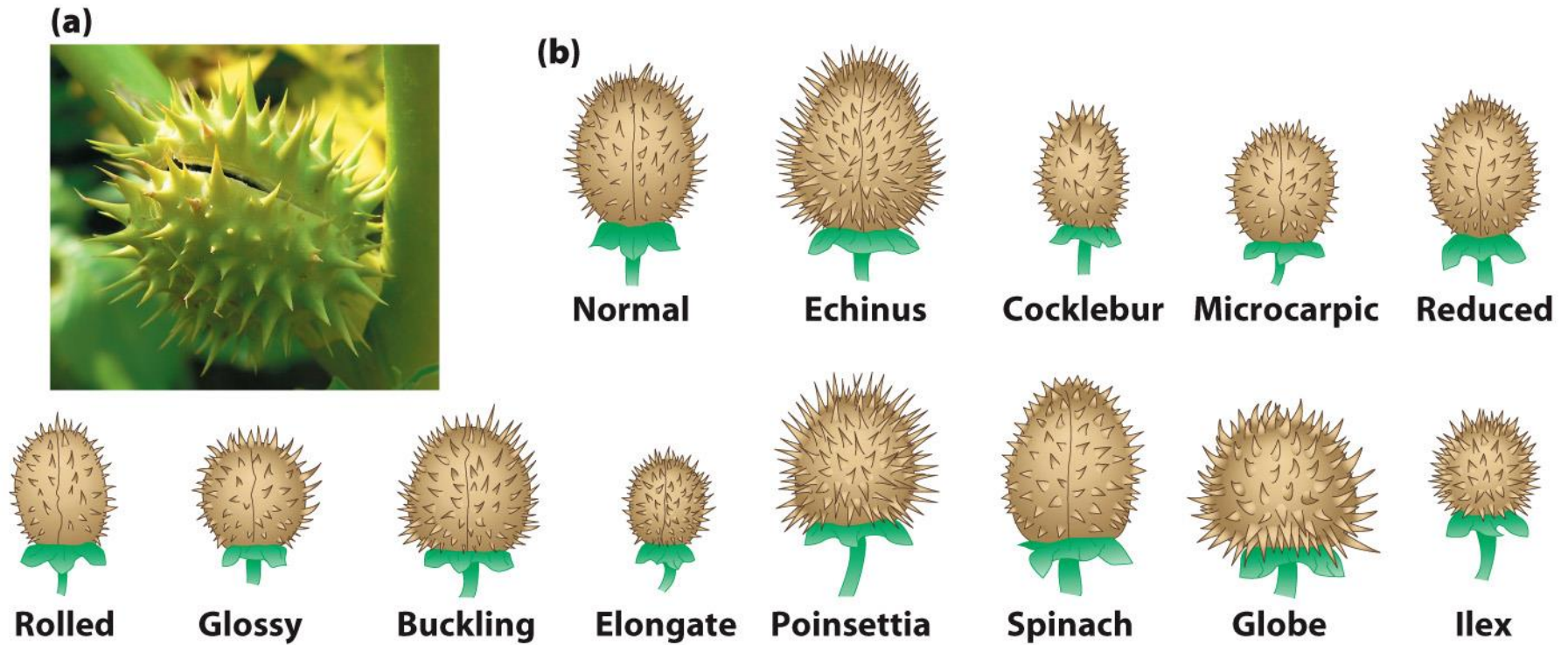


Figure 17-18

Introduction to Genetic Analysis, Eleventh Edition

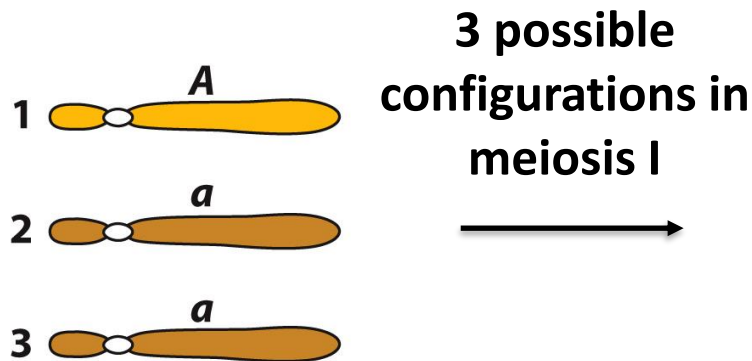
© 2015 W. H. Freeman and Company [Photo: iStockphoto/Thinkstock]

Think Break

Germline consequences of aneuploidy

Meiotic products of a trisomic

**3 possible meiosis outcomes
= 6 possible gametes**

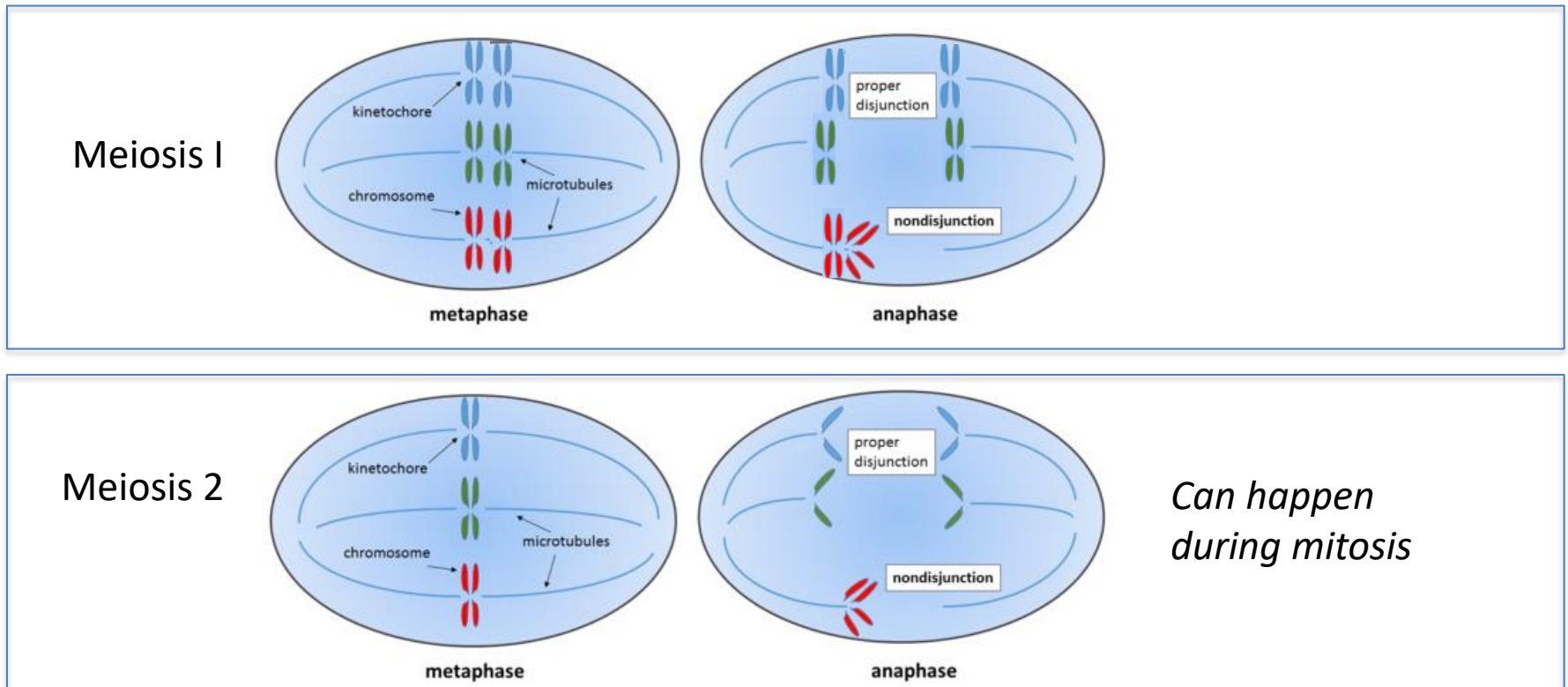


3 possible meiosis outcomes = 6 possible gametes	
One pole	Other pole
1  A 2  a	3  a
1  A 3  a	2  a
2  a 3  a	1  A

Figure 17-14
Introduction to Genetic Analysis, Eleventh Edition
© 2015 W. H. Freeman and Company

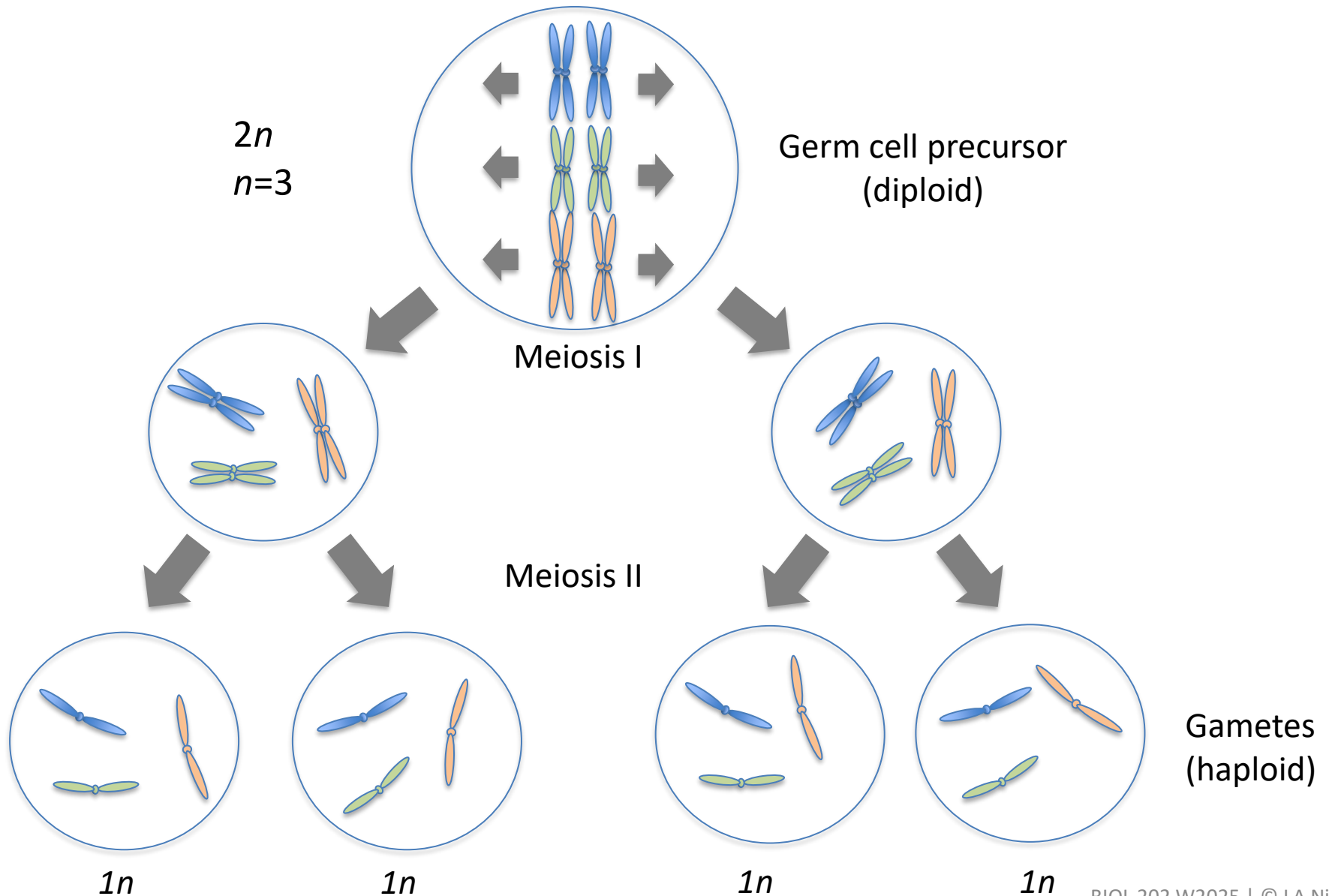
How does aneuploidy arise?

Aneuploidy can result from “non-disjunction” of homologous chromosomes during meiosis in diploid individuals

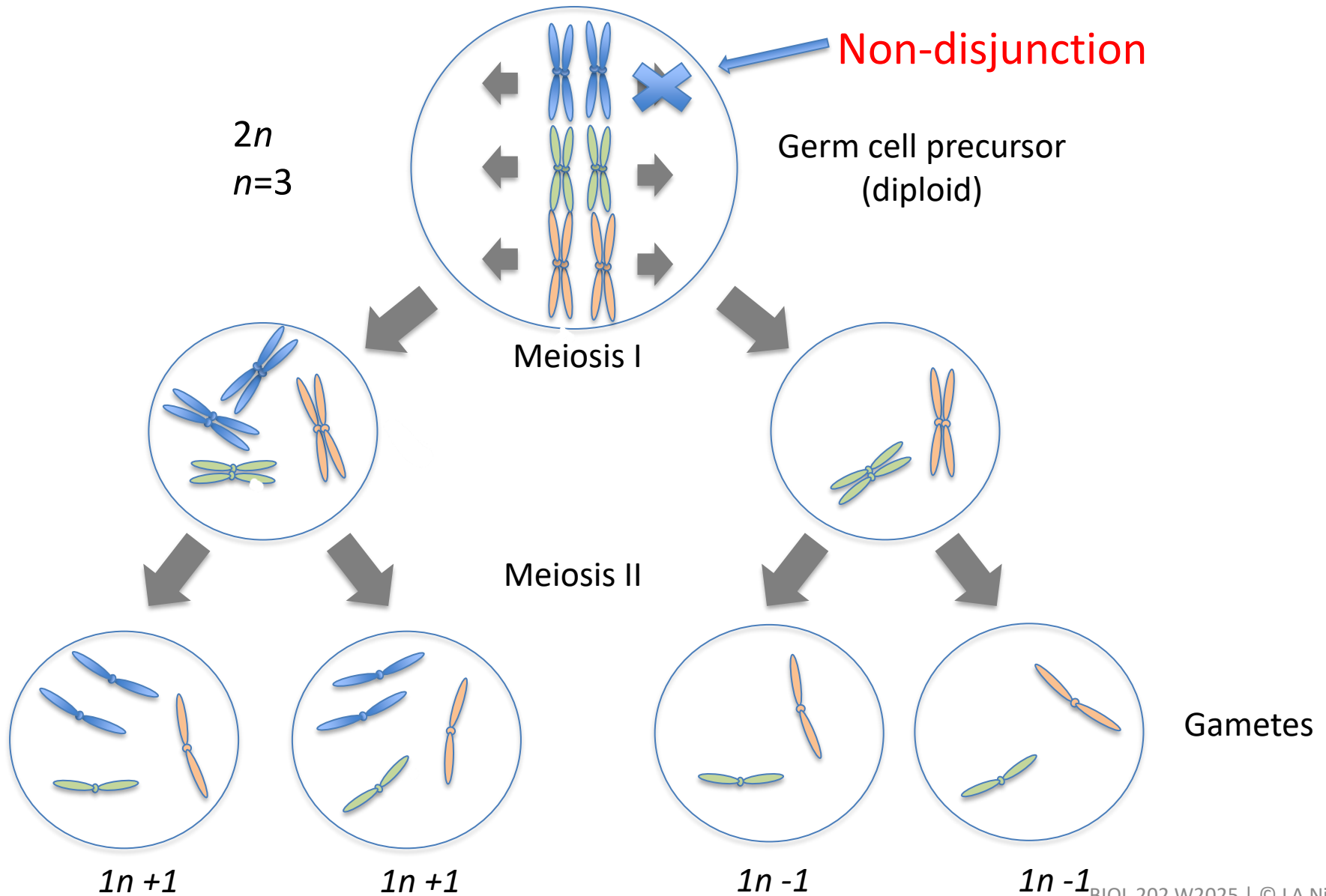


Nondisjunction occurs spontaneously. It is an example of a chance failure of a basic cellular process.

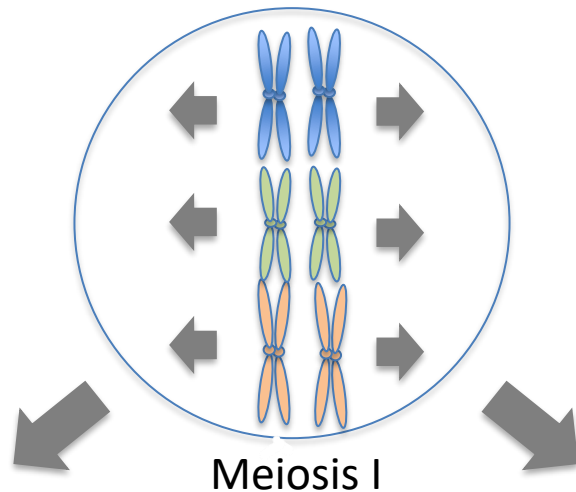
Normal meiosis in diploids



Non-disjunction during meiosis 1: 100% aneuploid gametes



Nondisjunction in meiosis I can reflect a failure in pairing/crossovers



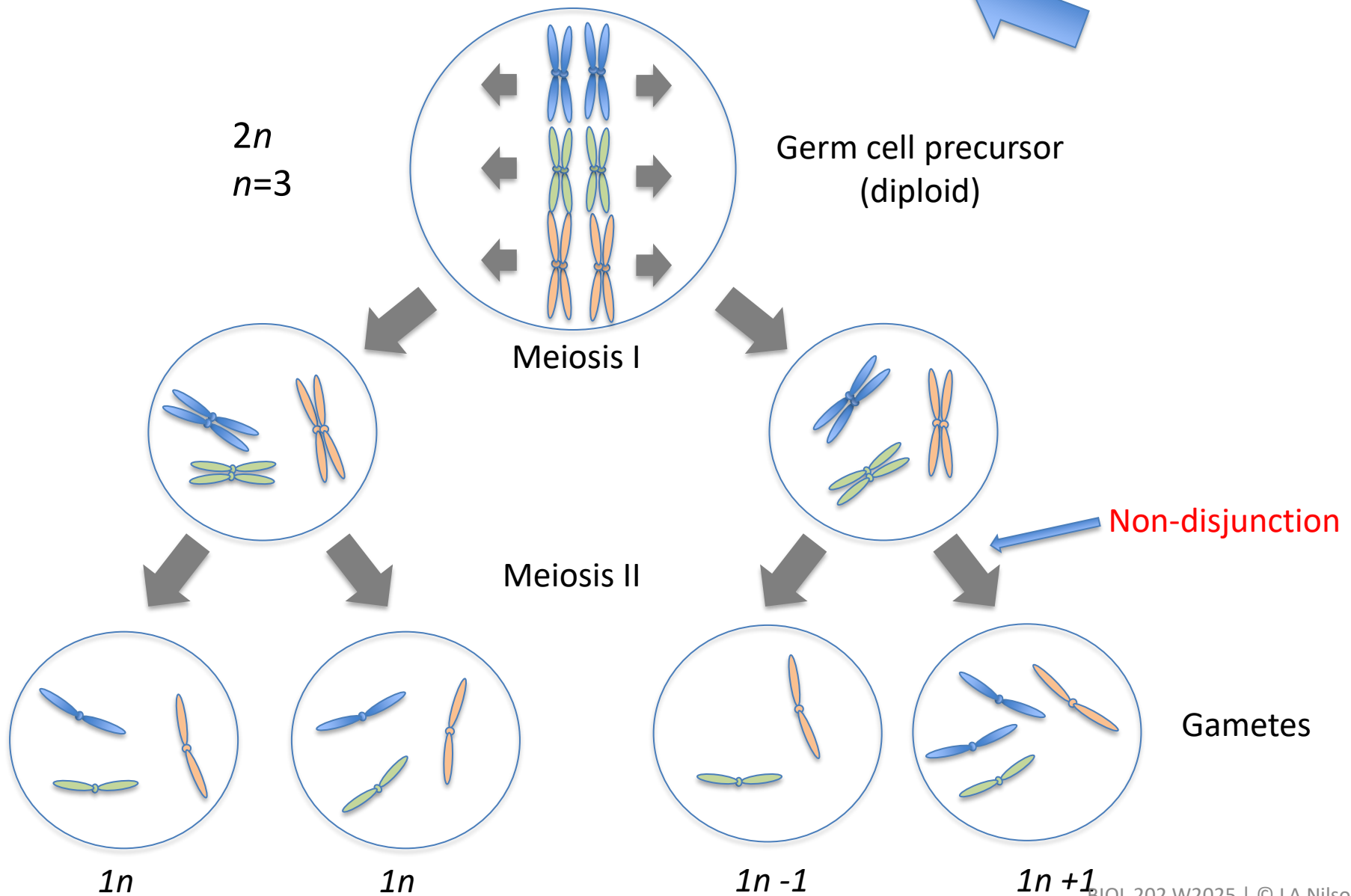
Nondisjunction is usually a rare failure of a basic cellular process.

Remember that crossing over (meiotic recombination) occurs during meiosis I, when homologs are paired.

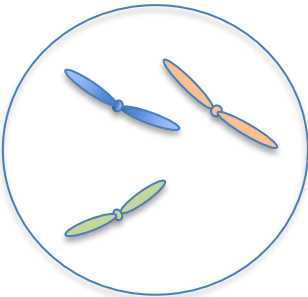
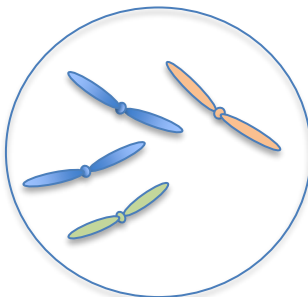
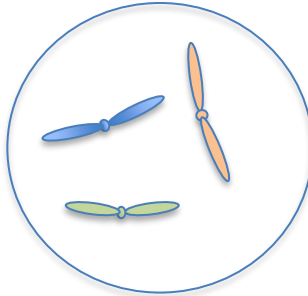
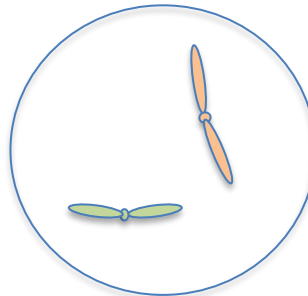
Some evidence that crossing over is required for normal nondisjunction:

1. In *Drosophila* (and human) trisomies the extra chromosome is non-recombinant (i.e. no crossovers).
2. Mutations that block crossing over also lead to more nondisjunction.

Non-disjunction during meiosis 2: 50% aneuploid gametes



Progeny produced by aneuploid gametes

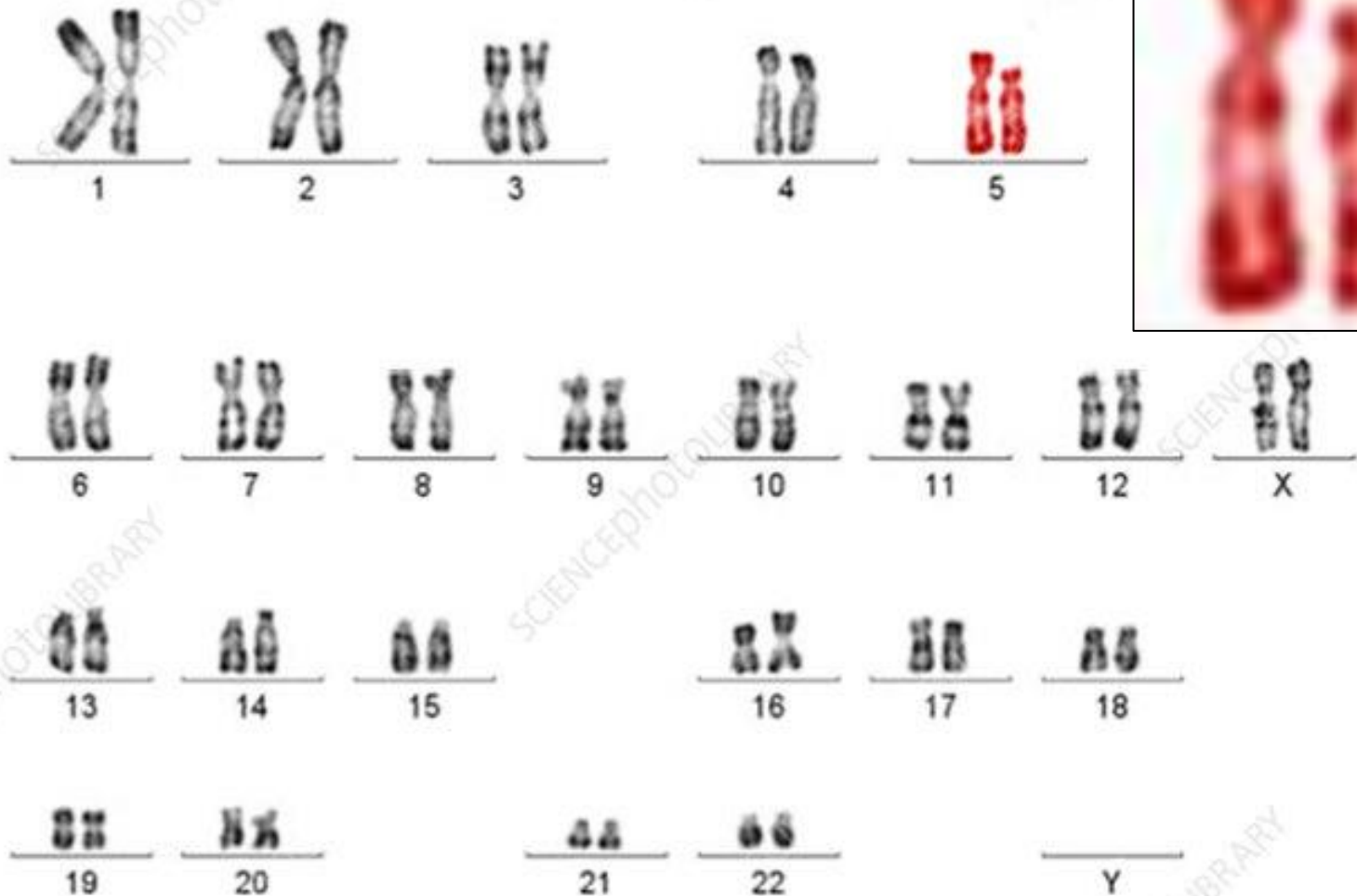
Gamete 1	Gamete 2	Zygote (progeny)
$1n$ 	$1n + 1$ 	
$1n$ 	$1n - 1$ 	

Where else have we seen the generation of
aneuploid gametes?

Large Scale Chromosomal Changes

Changes in Chromosome Structure

Cri-du-chat Syndrome



Cri-du-chat Syndrome

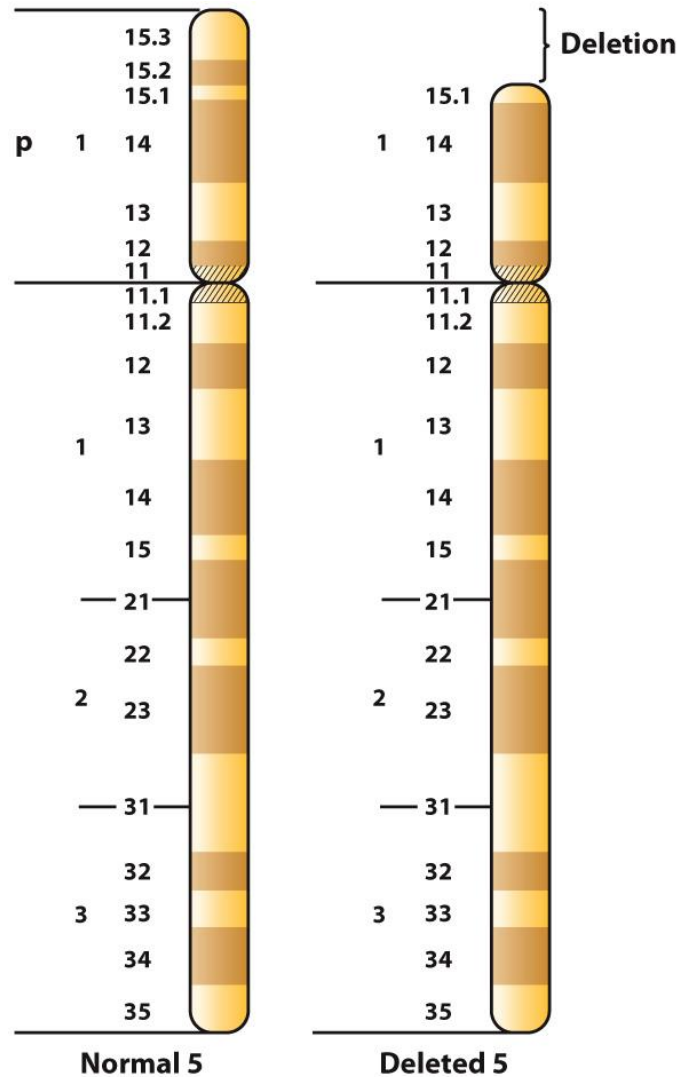
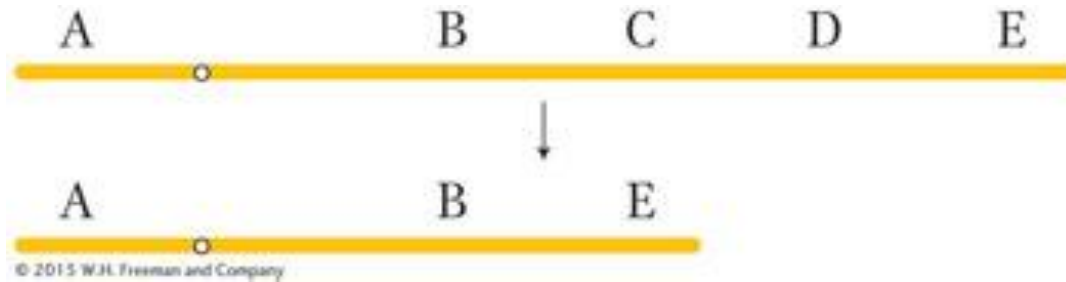


Figure 17-22
Introduction to Genetic Analysis, Eleventh Edition
© 2015 W. H. Freeman and Company

Chromosomal deletions

where do they come from?



Can be large
Can be small

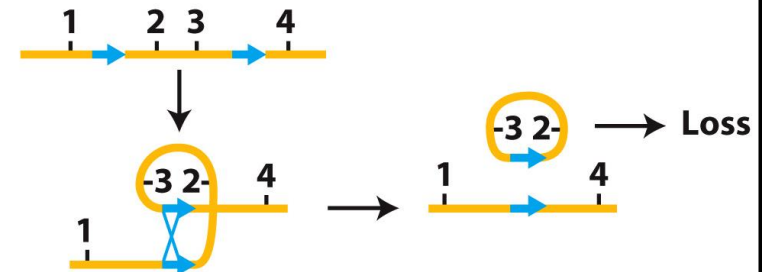
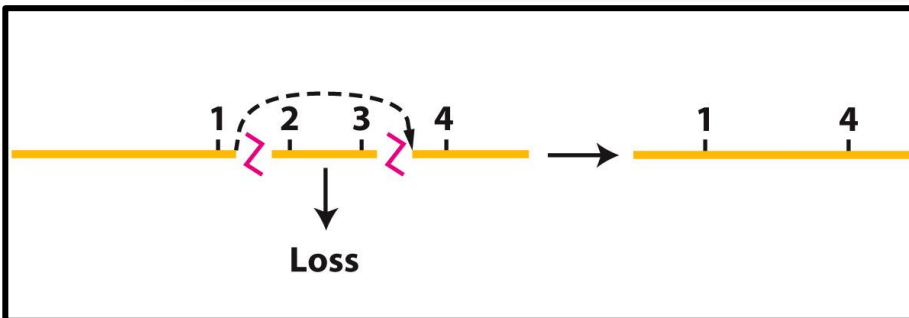
How would genes look on this diagram?

Tip: Make sure you understand the relative size of genes vs. chromosomes



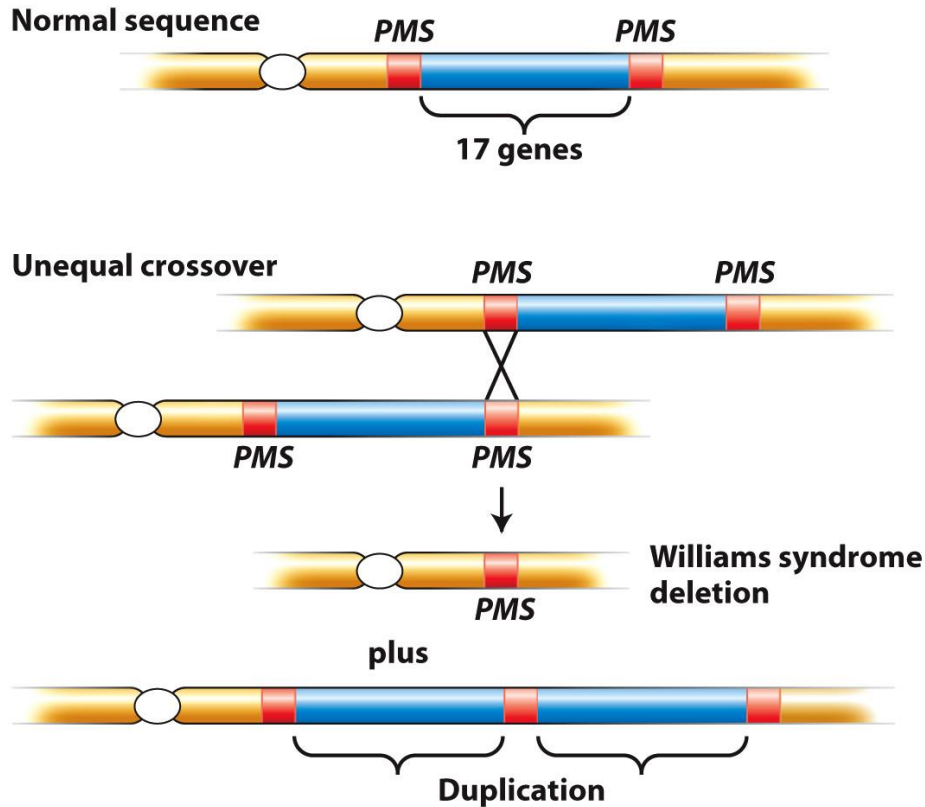
Breakage and rejoining

Crossing over between repetitive DNA



Chromosomal deletions

example: possible origin of the Williams syndrome deletion



Williams syndrome is found at a frequency of about 1 in 10,000 people.

Individuals often have pronounced musical or singing ability, as well as hyper-sociality.

The syndrome is almost always caused by a 1.5-Mb deletion on one homolog of chromosome 7, specifically at band 7q11.23.

Figure 17-23
Introduction to Genetic Analysis, Eleventh Edition
© 2015 W. H. Freeman and Company

Detecting chromosomal deletions

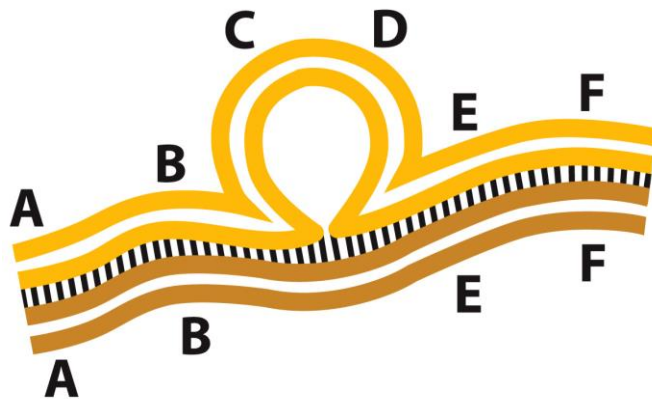
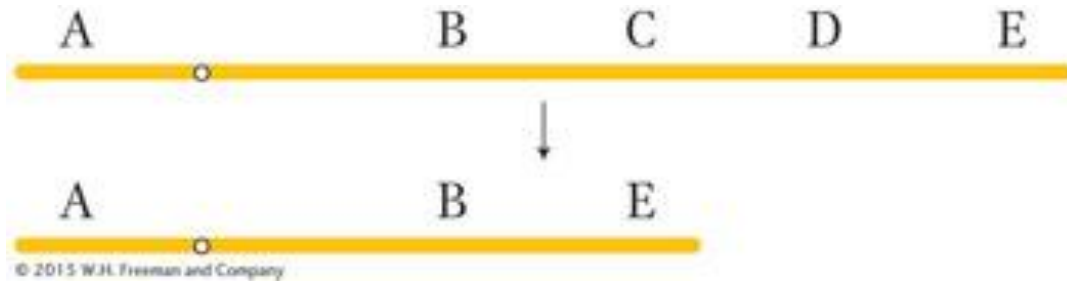


Figure 17-20a
Introduction to Genetic Analysis, Eleventh Edition
© 2015 W. H. Freeman and Company

Deletion loop observed in paired homologous chromosomes



Figure 17-20b
Introduction to Genetic Analysis, Eleventh Edition
William M. Gelbart, Harvard University

Deletion loops in *Drosophila* polytene chromosomes

Detecting chromosomal deletions

complementation test

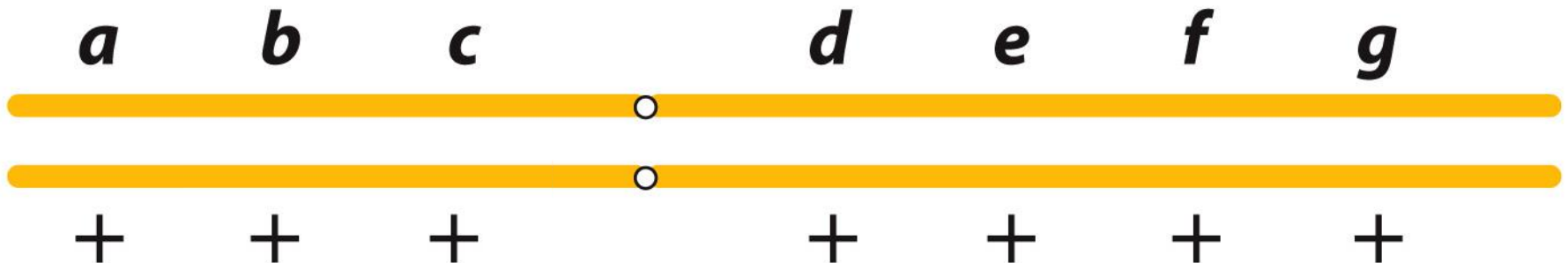
Wild type strain

"tester" strain with multiple
with recessive mutations

+++++++/++++++ x abcdefg/abcdefg



++++++/abcdefg **Phenotype?**



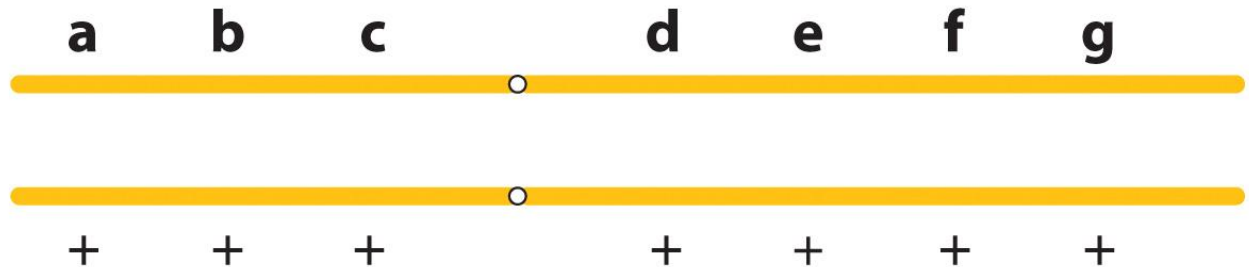
Detecting chromosomal deletions

complementation test

“tester” chromosome
with recessive mutations



Wild-type
chromosome



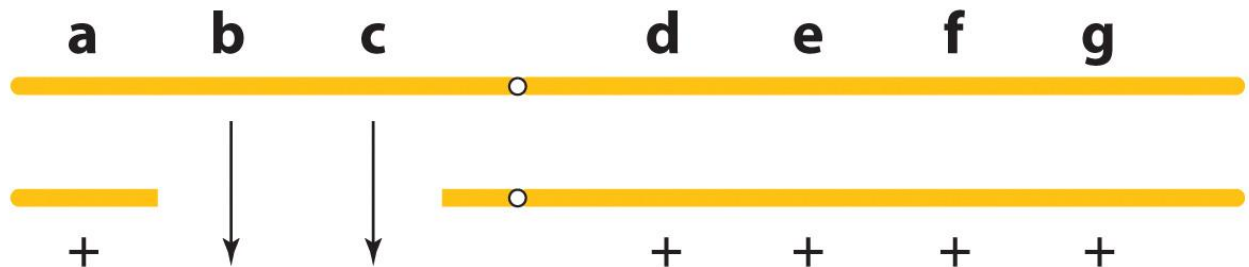
Phenotype

+ + + + + + + +

“tester” chromosome
with recessive mutations



Chromosome
with a deletion



Phenotype

+ b c + + + +

Detection of chromosomal deletions

Using a complementation test

“unknown strain” x abcdefg/abcdefg



“unknown strain” /abcdefg

Observed Phenotype: b c

Interpretation:

genes b and c are non-functional

a deletion removing both genes might be present

